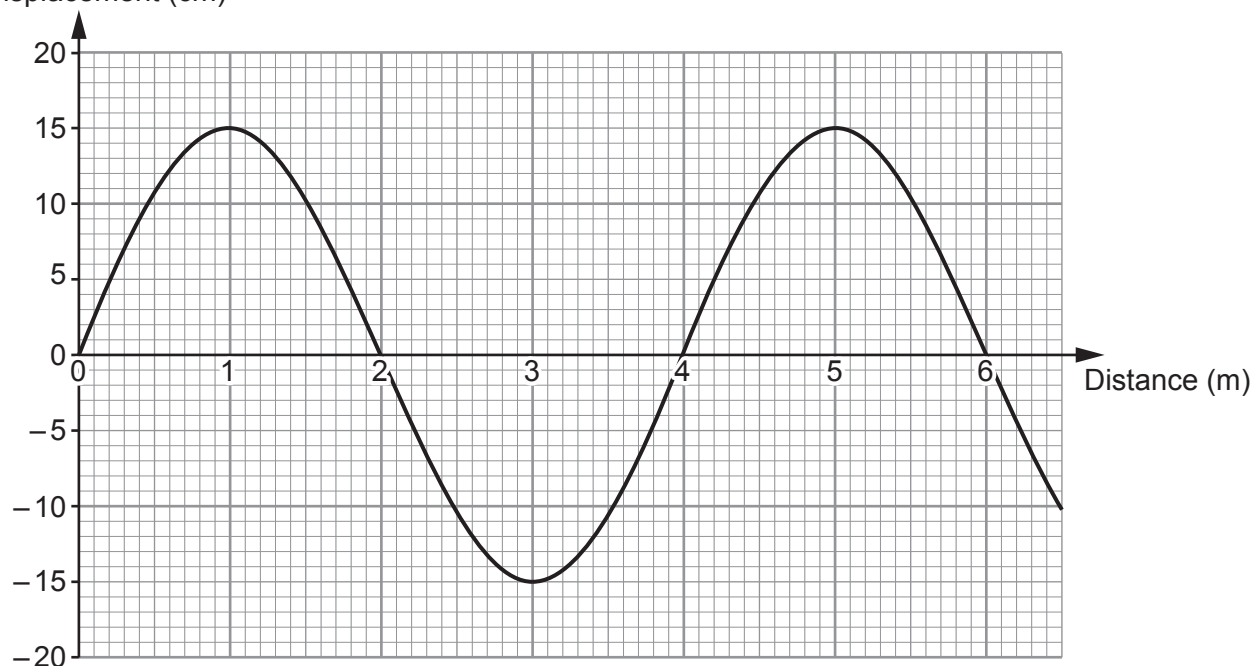


5. (a) The diagram shows a water wave.

Displacement (cm)



(i) State the wavelength of the wave. m [1]

(ii) The wave speed is 6 m/s.
Use the equation:

$$\text{frequency} = \frac{\text{wave speed}}{\text{wavelength}}$$

to determine the frequency of the wave.
Give the correct unit with your answer.

[3]

frequency =

unit =



Examiner
only

- (b) Students investigate the speed of water waves.
They measure the time taken for the waves to travel 100 cm in different depths of water.
Their results are shown below:

Depth of water (cm)	Time (s)			Mean time (s)	Mean wave speed (cm/s)
	Trial 1	Trial 2	Trial 3		
0.5	4.57	4.58	4.52	4.56	21.9
1.0	3.22	3.24	2.61	3.02	33.1
1.5	2.61	2.62	2.64	2.62	38.1
2.0	2.29	2.28	2.24	2.27	44.1
2.5	2.01	2.02	1.98	2.00	50.0

- (i) State the independent variable in this experiment. [1]

.....

- (ii) The students conclude that one of the time values they had recorded is anomalous.
Circle the anomalous result in the table. [1]

- (iii) Explain how you know that this result is anomalous. [1]

.....
.....

- (iv) Determine a more accurate value for the mean time for the depth in which there is an anomalous result. [2]

mean time = s

- (v) State a conclusion about the relationship between **depth of water** and **wave speed**. [1]

.....
.....



- (c) Water waves refract when their speed decreases at a boundary between deep and shallow water. Tick (✓) the box next to the diagram that correctly shows the refraction of these water waves. [1]

